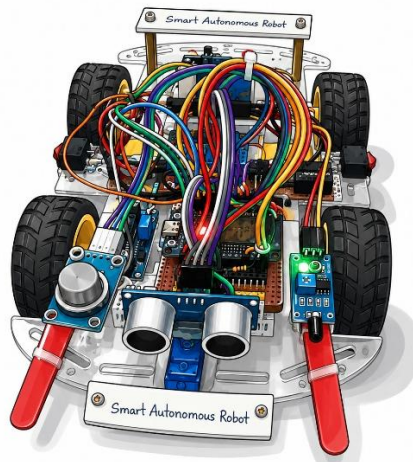


A FINAL PROJECT REPORT ON

SMART AUTONOMOUS FIRE AND GAS DETECTION ROBOT

An ESP32-Based Obstacle-Avoiding Robot with Real-Time Hazard Detection and Wi-Fi Monitoring



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August 15, 2026

Declaration

We hereby declare that this project report titled “Smart Autonomous Fire and Gas Detection Robot” documents a project that has been fully designed, constructed, and tested by us, under the supervision of Shefayatuj Johara Chowdhury. This report reflects the completed final version of the project. All sources of information used in this report have been duly acknowledged, and this work has not been submitted elsewhere for the award of any other degree or diploma.

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We extend our thanks to our families and friends for their continuous support and motivation during the development of this project.

Abstract

This report presents the design, implementation, and testing of a Smart Autonomous Fire and Gas Detection Robot built around the ESP32 microcontroller. The robot autonomously navigates its environment while avoiding obstacles and continuously monitors the surrounding air for hazardous gas concentrations and open flame using an MQ-2 gas sensor and a flame sensor respectively. Distance to obstacles is measured using an HC-SR04 ultrasonic sensor mounted on a servo motor, enabling the robot to scan left and right and choose a safe direction of travel. Locomotion is provided by four DC motors driven by an L298N motor driver.

On detecting a hazard, the robot immediately halts and raises a local alert through dedicated LEDs and a buzzer, while simultaneously updating a SH1106 OLED display with the current status. In addition, the ESP32 hosts a Wi-Fi-based web server that streams live sensor data including distance, gas concentration, fire status, temperature, and humidity to a browser-based dashboard, enabling remote, real-time monitoring from any device connected to the same network.

The system was developed and refined through an iterative testing process, addressing practical challenges such as collision avoidance, sensor calibration, and reliable turning, which are documented in detail within this report. The final system demonstrates reliable autonomous navigation, accurate hazard detection, and effective remote monitoring, making it a viable low-cost solution for safety patrol and hazard inspection applications.

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CHAPTER 1: INTRODUCTION

1.1 Background

Fires and gas leaks remain among the most common causes of property damage, injury, and loss of life in both residential and industrial settings. Early detection is critical: the sooner a hazard is identified, the greater the chance of preventing escalation. At the same time, manual inspection of large or hazardous areas is time-consuming and can expose personnel to unnecessary risk. Advances in low-cost microcontrollers, sensors, and wireless communication have made it feasible to build small autonomous robots that can patrol an area, continuously sense environmental conditions, and report findings in real time. This project explores the design and implementation of such a robot, combining autonomous navigation with fire and gas detection and remote monitoring over Wi-Fi.

1.2 Problem Statement

Existing safety monitoring in most small-scale environments relies on fixed-position smoke or gas detectors, which only respond to hazards within their immediate vicinity, and on manual human inspection, which is slow and potentially unsafe. There is a need for a mobile, autonomous system that can move through a space, detect hazards wherever they occur, avoid obstacles safely, and alert a remote observer immediately without requiring a human to be physically present at the point of danger.

1.3 Objectives

- To design and construct a four-wheel-drive autonomous robot chassis controlled by an ESP32 microcontroller.
- To implement reliable obstacle detection and avoidance using an ultrasonic sensor mounted on a servo motor.
- To detect hazardous gas concentrations and open flame using dedicated sensors (MQ-2 and flame sensor).
- To provide immediate local alerts through LEDs, a buzzer, and an OLED display upon hazard detection.
- To design and host a live WiFi web dashboard for remote, real-time monitoring of the robot's sensor data and status.
- To ensure smooth, safe, and reliable motion, including collision-free obstacle avoidance and dependable turning.

1.4 Scope of the Project

The scope of this project covers the design of the robot's hardware architecture, the development of firmware for sensor reading, decision-making, and motor control, and the implementation of a Wi-Fi-based dashboard for remote monitoring. The project focuses on indoor, short-range operations on flat or moderately uneven surfaces. It does not cover long-range outdoor navigation, GPS-based positioning, or integration with external fire-suppression systems, which are identified as potential directions for future work.

1.5 Report Organization

This report is organized into six chapters. Chapter 1 introduces the project background, problem statement, and objectives. Chapter 2 reviews existing related systems and identifies gaps this project addresses. Chapter 3 describes the system design and methodology, including the hardware architecture and working principle. Chapter 4 details the implementation of the software, including the obstacle avoidance algorithm, hazard detection logic, and web dashboard. Chapter 5 presents the testing process, issues encountered, and results obtained. Chapter 6 concludes the report and outlines directions for future work.

CHAPTER 2: LITERATURE REVIEW

2.1 Existing Systems

Fixed smoke and gas detectors are the most common form of hazard detection currently deployed in homes and industrial facilities. These devices are inexpensive and reliable within their coverage area but are limited to a single, fixed location. Separately, obstacle-avoiding robots using ultrasonic or infrared sensors have been widely studied and implemented for applications ranging from educational robotics to warehouse automation, typically relying on simple threshold-based distance checks to trigger a turn. More recent IoT-enabled safety systems combine sensors with Wi-Fi or GSM modules to send alerts to a remote server or mobile application when a hazard is detected, improving response time compared to purely local alarms.

2.2 Limitations of Existing Systems

- Fixed detectors cannot cover an entire area unless installed in large numbers, increasing cost.
- Most low-cost obstacle-avoiding robots rely on simple, single-attempt turning logic that performs poorly when the chassis uses gear motors, or when obstacles surround the robot on multiple sides.
- Many existing hazard-alert systems handle only one hazard type at a time and do not clearly represent simultaneous, multiple hazards to the observer.
- Gas sensor readings are highly sensitive to the specific sensor unit and ambient conditions; systems using a fixed detection threshold are prone to false positives or false negatives.

2.3 Proposed Improvements

This project addresses the above limitations by combining mobile hazard detection with autonomous navigation, so a single low-cost robot can cover an extended area rather than a fixed point. The obstacle avoidance logic incorporates active braking, smooth pivot turning suited to gear-motor-driven chassis, and a back-up-and-retry strategy for situations where the robot is boxed in. The hazard detection system evaluates fire and gas independently every cycle, so simultaneous hazards are both reported rather than one masking the other, and the gas sensor is automatically calibrated to its own clean-air baseline at startup to reduce false detections. Finally, a live Wi-Fi dashboard provides continuous remote visibility without requiring additional infrastructure beyond a local Wi-Fi network.

CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY

3.1 System Architecture

The system is organized into three functional groups: input devices (sensors), a central processing unit (the ESP32 microcontroller), and output devices (actuators and indicators). The ESP32 continuously reads all sensors, executes the navigation and safety decision logic, and drives the outputs accordingly. Figure 3.1 illustrates the overall system architecture.

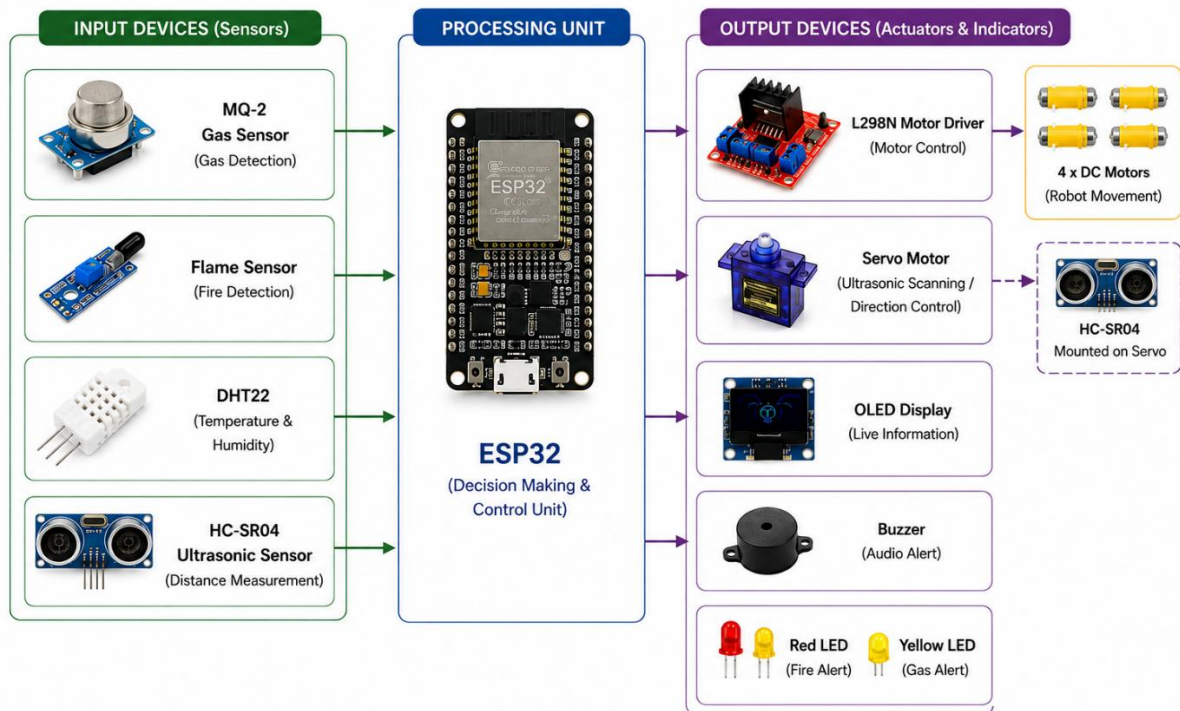


Figure 3.1: System Architecture of the Completed Robot

3.2 Hardware Components

This section describes each hardware component used in the construction of the robot, along with its role in the overall system. A placeholder is provided beneath each description for a photograph of the actual component used.

3.2.1 ESP32 DevKit (Microcontroller)

The ESP32 is a low-cost, dual-core microcontroller with built-in Wi-Fi and Bluetooth connectivity and a rich set of GPIO, ADC, and PWM-capable pins. It serves as the central processing and control unit of the robot, responsible for reading all sensors, executing the

navigation and hazard-detection logic, driving the motors and indicators, and hosting the Wi-Fi web dashboard.

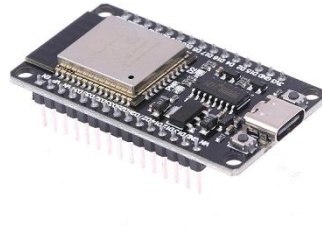


Figure 3.2: ESP32 DevKit

3.2.2 MQ-2 Gas Sensor

The MQ-2 is a semiconductor-type gas sensor sensitive to combustible and toxic gases such as LPG, methane, and smoke. It produces an analog output voltage proportional to gas concentration. In this project, its reading is auto calibrated against ambient air at startup and used to detect hazardous gas levels.



Figure 3.3: MQ-2 Gas Sensor

3.2.3 Flame Sensor

The flame sensor module detects infrared radiation in the wavelength range typically emitted by fire (roughly 760–1100 nm) and produces a digital output that goes low when flame is detected. It provides the robot's primary means of fire detection.



Figure 3.4: Flame Sensor

3.2.4 DHT22 Temperature and Humidity Sensor

DHT22 is a digital sensor that measures ambient temperature and relative humidity, communicating with the microcontroller over a single-wire digital interface. It provides supplementary environmental data displayed on the OLED and web dashboard.



Figure 3.5: DHT22 Temperature and Humidity Sensor

3.2.5 HC-SR04 Ultrasonic Sensor

The HC-SR04 measures distance by emitting an ultrasonic pulse and timing the echo reflected back from a nearby object, providing a usable range of approximately 2 cm to 400 cm. Mounted on the servo motor, it allows the robot to measure distance directly ahead and scan left and right to assess available clearance.



Figure 3.6: HC-SR04 Ultrasonic Sensor

3.2.6 SG90 Servo Motor

The SG90 is a small, lightweight servo motor capable of rotating to a commanded angle within an approximately 180-degree range. It is used to rotate the ultrasonic sensor so the robot can scan for obstacles to its left and right.



Figure 3.7: SG90 Servo Motor

3.2.7 L298N Motor Driver

The L298N is a dual H-bridge motor driver capable of independently controlling the speed and direction of two DC motor channels. It receives direction and PWM speed signals from the ESP32 and supplies the higher current required to drive the robot's four DC motors.



Figure 3.8: L298N Motor Driver

3.2.8 SH1106 OLED Display

The SH1106 is a 0.96-inch monochrome OLED display communicating over the I2C bus. Because each pixel is self-emissive, it offers high contrast and low power consumption. It is used to show the robot's live status, sensor readings, and Wi-Fi dashboard address.



Figure 3.9: SH1106 OLED Display

3.2.9 Buzzer and LED Indicators

A piezoelectric buzzer, a red LED, and a yellow LED provide immediate local alerts. The red LED blinks together with the buzzer during a fire alert, and the yellow LED blinks together with the buzzer during a gas alert; both blink together if both hazards are detected simultaneously.



Figure 3.10: Buzzer and LED Indicators

3.2.10 Li-ion Battery Pack

A rechargeable Li-ion battery pack (7.4V / 11.1V) serves as the robot's primary power source, regulated down to 5V and 3.3V as required by the ESP32, sensors, and other onboard electronics.



Figure 3.11: Li-ion Battery Pack

3.3 Pin Connection Details

Table 3.1 and Table 3.2 present the complete pin mapping between each component and the ESP32, as implemented in the firmware.

Table 3.1: Sensor Pin Connections

Component	Pin	ESP32 Pin
MQ-2 Gas Sensor	VCC / GND / A0	5V / GND / GPIO 34
Flame Sensor	VCC / GND / D0	3.3V / GND / GPIO 35
DHT22	VCC / GND / DATA	3.3V / GND / GPIO 27
HC-SR04 Ultrasonic	VCC / GND / TRIG / ECHO	5V / GND / GPIO 5 / GPIO 18

Table 3.2: Actuator and Display Pin Connections

Component	Pin	ESP32 Pin
Servo Motor (SG90)	Signal / VCC / GND	GPIO 13 / 5V / GND
L298N Motor Driver	ENA / IN1 / IN2	GPIO 14 / GPIO 26 / GPIO 25
L298N Motor Driver	ENB / IN3 / IN4	GPIO 33 / GPIO 32 / GPIO 15
SH1106 OLED (I2C)	VCC / GND / SDA / SCL	3.3V / GND / GPIO 21 / GPIO 22
Buzzer	+ / -	GPIO 12 / GND
Red LED (Fire Alert)	+ / -	GPIO 4 (330Ω resistor) / GND
Yellow LED (Gas Alert)	+ / -	GPIO 2 (330Ω resistor) / GND

Note: All component grounds (ESP32, L298N, battery) share a common ground reference. GPIO 34 and 35 are input-only pins on the ESP32, which is compatible with the analog gas sensor and digital flame sensor used in this design.

3.4 Working Principle

On power-up, the ESP32 initializes all connected sensors, establishes a Wi-Fi connection, and performs an automatic calibration of the gas sensor against ambient air. The robot then enters a continuous control loop, illustrated in Figure 3.12, in which fire and gas conditions are checked first, followed by obstacle detection and forward navigation. The OLED display and web dashboard are updated on every iteration of the loop.

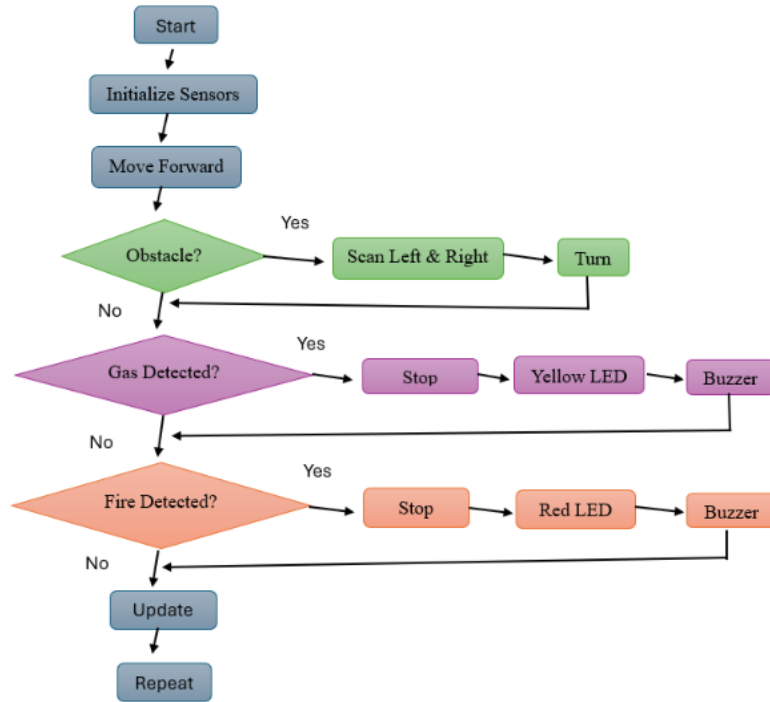


Figure 3.12: Main Control-Loop Flowchart

The decision priority implemented in the control loop is as follows:

- If fire is detected, the robot stops, the red LED blinks together with the buzzer, and the dashboard displays a fire alert.
- If gas is detected, the robot stops, the yellow LED blinks together with the buzzer, and the dashboard displays a gas alert.
- If both fire and gas are detected simultaneously, both LEDs blink together and the dashboard displays a combined alert.
- If no hazard is present, the robot moves forward while continuously scanning for obstacles.

3.5 Final Prototype

Figure 3.13 shows a photograph of the fully assembled and completed robot prototype, built and tested according to the design described in this chapter.

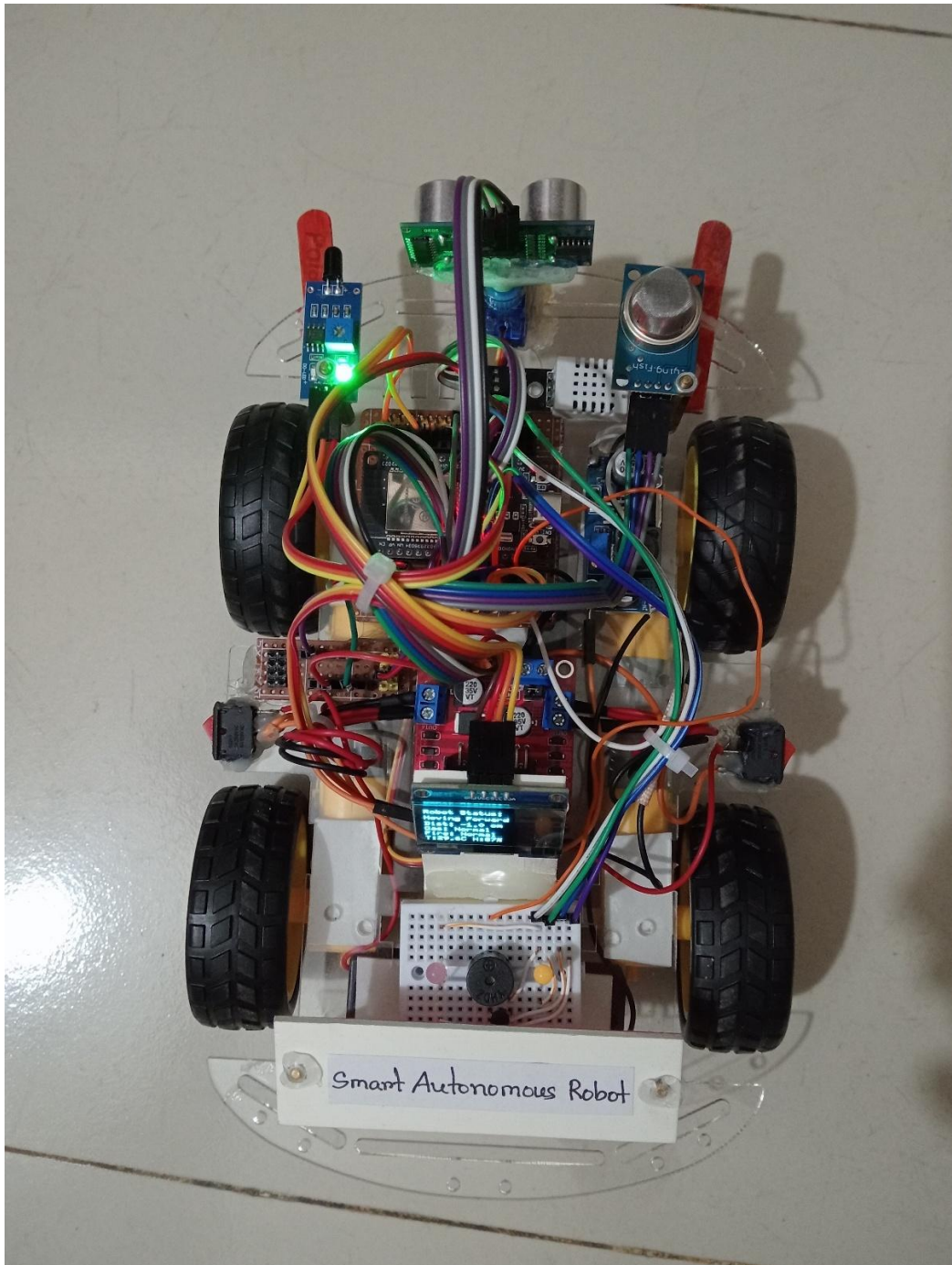


Figure 3.13: Photograph of the Completed Robot Prototype

CHAPTER 4: IMPLEMENTATION

4.1 Software Development Environment

The firmware was developed in C++ using the Arduino framework for the ESP32 platform. The following libraries were used: ESP32Servo for servo control, the Adafruit DHT sensor library for temperature and humidity readings, U8g2 for driving the SH1106 OLED display, and the built-in Wi-Fi and WebServer libraries for hosting the live dashboard. The firmware is structured as a set of independent, non-blocking routines coordinated from the main loop, ensuring that Wi-Fi requests are served promptly even while the robot is moving, turning, or braking.

4.2 Obstacle Avoidance Algorithm

The HC-SR04 ultrasonic sensor, mounted on the servo motor, measures the distance directly ahead of the robot at regular intervals. When an obstacle is detected within the pre-defined safe threshold, the following sequence is executed:

- Active braking is applied rather than simply cutting motor power, the H-bridge briefly shorts the motor windings, bringing the robot to a fast stop instead of coasting forward on momentum.
- The servo sweeps the ultrasonic sensor to the left and right to measure the available clearance on each side.
- If one side has significantly more clearance than the other, the robot performs a pivot turn (one side of the wheels driven forward, the other in reverse) toward the more open side.
- If the left and right clearances are nearly equal, the robot alternates its turn direction between attempts, avoiding a persistent bias toward one side.
- If both sides are blocked, the robot reverses briefly to create additional space before re-scanning and attempting the turn again.

Forward motion uses a gradual speed ramp rather than an immediate jump to full power, and turns are similarly ramped up and down, resulting in smoother, less jerky motion overall.

4.3 Hazard Detection and Alert System

Gas concentration is read from the MQ-2 sensor's analog output. Because MQ-2 sensors require a warm-up period and exhibit unit-to-unit variation in baseline reading, the firmware performs an automatic calibration routine at startup: after a warm-up delay, it samples the sensor

repeatedly to establish the ambient “clean air” baseline, and sets the detection threshold relative to this baseline rather than using a fixed value.

The flame sensor's digital output is polled continuously to detect the presence of fire. Fire and gas conditions are evaluated independently on every loop iteration, so that both can be active and reported simultaneously. A single shared timer drives the red LED, yellow LED, and buzzer, ensuring that fire-only, gas-only, and combined fire-and-gas alerts all blink correctly and remain synchronized.

4.4 Wi-Fi Web Dashboard

The ESP32 hosts a lightweight web server that serves an HTML dashboard directly from the device, accessible from any browser on the same Wi-Fi network via the robot's IP address (shown on the OLED display at startup). The dashboard polls a JSON data endpoint once per second and displays:

- The robot's current operating state (e.g. moving forward, scanning for an obstacle, or an active hazard alert), with a pulsing visual indicator during alerts.
- Independent fire and gas status indicators, so that simultaneous alerts are both clearly visible.
- Live distance and gas-level readings, each shown with a proportional progress bar.
- Live temperature and humidity readings.
- A connection-status indicator that reflects whether the dashboard is currently receiving data from the robot.

Figures 4.1 to 4.3 present screenshots of the web dashboard captured during operation, illustrating the normal operating view, an active hazard alert, and the live sensor readings respectively.



Figure 4.1: Web Dashboard — Normal Operation

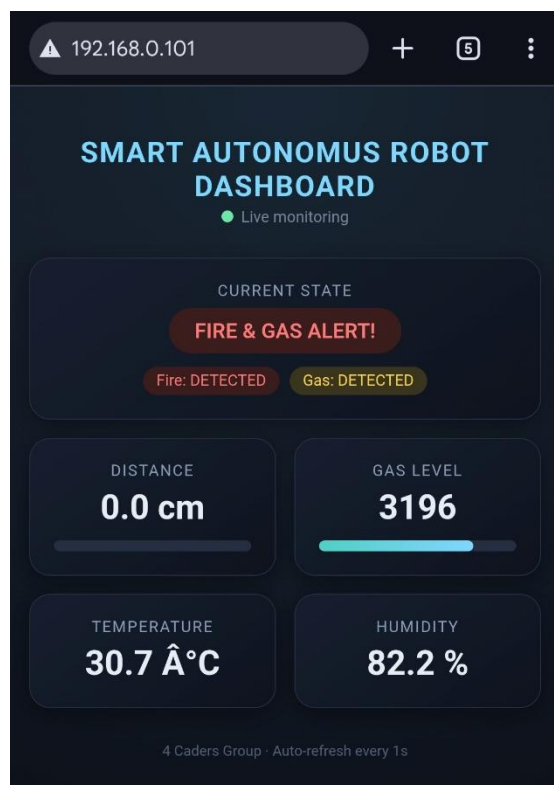


Figure 4.2: Web Dashboard — Hazard Alert Active

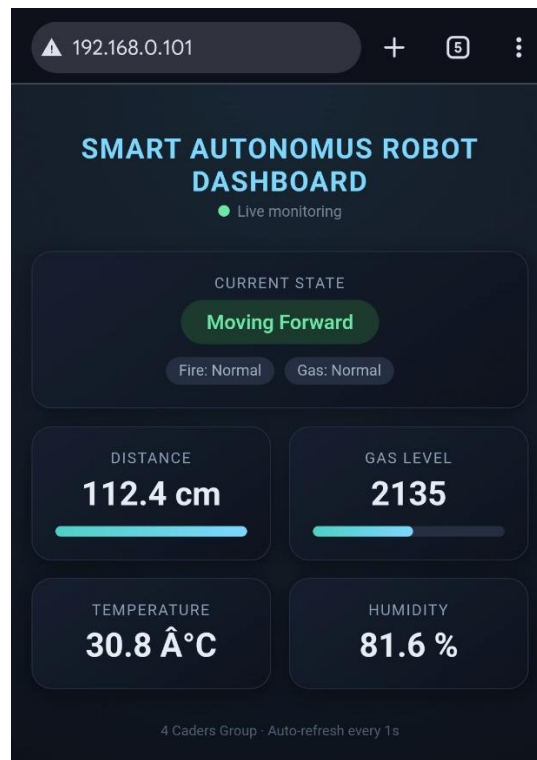


Figure 4.3: Web Dashboard — Live Sensor Readings

CHAPTER 5: TESTING, RESULTS AND DISCUSSION

5.1 Development Iterations and Issue Resolution

The system was refined through several rounds of bench testing. Table 5.1 summarizes the key issues identified during testing, their root causes, and the corresponding resolutions implemented in the firmware.

Table 5.1: Development Issues and Resolutions

Issue Observed	Root Cause	Resolution
Robot collided with obstacles before stopping	Stop distance too small; motor power cut but robot coasted forward on momentum	Increased the safe stop distance, added an emergency-stop distance, and switched to active braking instead of simply cutting power
Gas sensor showed “detected” even in normal air	MQ-2 requires a warm-up period and has a unit-specific baseline; a fixed threshold did not match the sensor	Added an automatic warm-up and calibration routine that measures the sensor's own clean-air baseline at startup
Dashboard displayed only one alert even when both were active	The control logic returned early after checking fire, so gas was never evaluated in the same cycle	Restructured the logic to evaluate fire and gas independently every cycle, using a shared blink timer for both LEDs
Robot consistently turned in the same direction when both sides were open	Turn direction always defaulted to whichever side measured marginally larger	Introduced a tie-break margin; when both sides are roughly equal, the turn direction now alternates between attempts
Turning was jerky, or the robot failed to turn at all	A single-side “push” turn was used, but the gear motors do not free-wheel, causing the idle side to drag	Replaced this with a proper pivot turn (one side forward, one side reverse) using a smooth speed ramp, with a reverse-and-retry step for when both sides are blocked

5.2 Results

- The robot reliably detects obstacles and comes to a complete stop without collision under normal test conditions.
- Fire and gas hazards are detected and alerted both locally (LEDs, buzzer, OLED) and remotely (web dashboard), including cases where both hazards occur at the same time.
- Turning behaviour is smooth and reliable, including automatic recovery when the robot is boxed in on both sides.
- The web dashboard accurately reflects the robot's real-time sensor readings and operating state, updating once per second.

5.3 Discussion

The results indicate that the combination of active braking, gradual speed ramping, and true pivot turning is essential for reliable navigation on chassis driven by gear motors, as single side “push” turning proved insufficient in practice. Similarly, sensor-specific calibration was found to be necessary for consistent gas detection, given the variability observed between individual MQ-2 units. The independent evaluation of fire and gas conditions on every control cycle was an important correction, as sequential, early-return logic can silently suppress the reporting of a second, simultaneous hazard a failure mode that would be particularly undesirable in a safety-oriented system.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project has successfully demonstrated the design, implementation, and testing of a Smart Autonomous Fire and Gas Detection Robot. The system integrates autonomous obstacle avoidance, dual hazard detection, local alerting, and remote monitoring over Wi-Fi into a single low-cost platform built around the ESP32 microcontroller. Through iterative testing, key practical challenges including collision prevention, sensor calibration, and reliable turning on a gear-motor-driven chassis were identified and resolved, resulting in a system that operates safely, detects hazards accurately, and provides clear, real-time visibility through its web dashboard.

6.2 Future Scope

- Incorporate wheel encoders for precise, distance-based turning rather than timed turns.
- Integrate a camera module to enable visual obstacle recognition and hazard confirmation.
- Add historical data logging and trend charts to the web dashboard.
- Implement remote manual-override control accessible from the dashboard.
- Add mobile push notifications for immediate hazard alerts.
- Extend the sensor suite to detect additional hazards, such as carbon monoxide or structural smoke density.

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APPENDIX A: PIN CONNECTION SUMMARY

This appendix reproduces the complete pin connection tables from Chapter 3 for quick reference.

Sensors

Component	Pin	ESP32 Pin
MQ-2 Gas Sensor	VCC / GND / A0	5V / GND / GPIO 34
Flame Sensor	VCC / GND / D0	3.3V / GND / GPIO 35
DHT22	VCC / GND / DATA	3.3V / GND / GPIO 27
HC-SR04 Ultrasonic	VCC / GND / TRIG / ECHO	5V / GND / GPIO 5 / GPIO 18

Actuators and Display

Component	Pin	ESP32 Pin
Servo Motor (SG90)	Signal / VCC / GND	GPIO 13 / 5V / GND
L298N Motor Driver	ENA / IN1 / IN2	GPIO 14 / GPIO 26 / GPIO 25
L298N Motor Driver	ENB / IN3 / IN4	GPIO 33 / GPIO 32 / GPIO 15
SH1106 OLED (I2C)	VCC / GND / SDA / SCL	3.3V / GND / GPIO 21 / GPIO 22
Buzzer	+ / -	GPIO 12 / GND
Red LED (Fire Alert)	+ / -	GPIO 4 (330Ω resistor) / GND
Yellow LED (Gas Alert)	+ / -	GPIO 2 (330Ω resistor) / GND